

Quick Installation (with a Ready-to-Use SD Card)

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This quick start guide sets up the TurtleBot3 and the PC for line following, using a camera with the `v4l2_camera` ROS package.

It follows the [official manual](#) to install ROS Humble, but offers a much quicker alternative to set up the TurtleBot3 Raspberry Pi (RPI): a ready-to-use SD card image.

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1. PC installation

Follow the official Robotis quick start guide to set up a PC with Ubuntu 22.04 and ROS 2 Humble:

- [quick_start_guide/pc_setup](#)

Note: at the corresponding step, install both ROS 2 Humble Desktop and the ROS development tools:

```
sudo apt install ros-humble-desktop
sudo apt install ros-dev-tools
```

Then install a few useful tools:

- `rpi-imager`, to write the SD card:

```
sudo apt install rpi-imager
```

- `terminator`, to open several terminals in a split view:

```
sudo apt install terminator
```

- `codium` (VSCodium, the open-source build of VS Code), to edit Python code:

```
sudo snap install codium --classic
```

Then run `codium` and install the Python extension (`ms-python.python`) for advanced editing, live syntax checking, debugging, etc.

2. TurtleBot3 RPi installation (SBC and OpenCR setup)

There are two ways to install the TurtleBot3 Raspberry Pi (the SBC, single board computer):

- [Option A](#): write a ready-to-use SD card image (recommended, much quicker),
- [Option B](#): follow the step-by-step instructions of the Robotis manual.

Option A: with a ready-to-use SD card image

As an alternative to building the RPi system step by step as in the Robotis manual ([quick_start_guide/sbc_setup](#)), a ready-to-use SD card image is available. It contains Ubuntu 22.04, ROS 2 Humble, the TurtleBot3 packages, the legacy camera driver and the camera tools (V4L2):

Download: [turtlebot3_humble_v4l2camera.img.gz](#)

MD5 checksum: `17adbbfc7703a4f99a9bbc24980356a7`
`turtlebot3_humble_v4l2camera.img.gz`

Write the image to the SD card

1. Check that the download is not corrupted: the following command must display the checksum above.

```
md5sum turtlebot3_humble_v4l2camera.img.gz
```

2. Decompress the image (about 7.5 GB once decompressed):

```
gunzip -k turtlebot3_humble_v4l2camera.img.gz
```

3. Write the image to the SD card with `rpi-imager`:
 - in the Raspberry Pi Imager window, choose *Operating System* > *Use custom* and select the decompressed file `turtlebot3_humble_v4l2camera.img`,
 - choose your SD card as *Storage* and write the image,
 - no OS customization is needed.

Network configuration

The TurtleBot3 network (Wi-Fi or wired) is configured in the file `/etc/netplan/50-cloud-init.yaml`. It can be edited either on the TurtleBot3 itself, with a screen and a keyboard (method 1), or directly on the SD card from the PC (method 2).

Method 1: with an HDMI screen and a USB keyboard

- As detailed in [Configure the Raspberry Pi](#):

- insert the SD card into the RPi,
- connect a screen to the RPi with an HDMI cable, and connect a USB keyboard,
- power on the TurtleBot3 (**after** connecting the HDMI cable!).
- Log in with the user `ubuntu` and the password `turtlebot`.
- The directory `/etc/netplan` contains 1 regular file and 2 hidden files:

```
ls -A /etc/netplan
```

```
50-cloud-init.yaml  .50-cloud-init.yaml.dhcp  .50-cloud-init.yaml.static
```

The two hidden files are ready-made configurations. Copy the one you need over `50-cloud-init.yaml`, then edit it:

- **a) DHCP:** your Wi-Fi hotspot/router has a DHCP server that automatically gives an IP address to the TurtleBot3.

```
cd /etc/netplan
sudo cp .50-cloud-init.yaml.dhcp 50-cloud-init.yaml
```

Then edit the file: replace `c138proj` with your Wi-Fi SSID and `password_here` with your Wi-Fi password.

```
sudo nano 50-cloud-init.yaml
```

- **b) Static IP:** there is no DHCP server, or you use a wired connection (`eth0`).

```
cd /etc/netplan
sudo cp .50-cloud-init.yaml.static 50-cloud-init.yaml
```

Then edit the file: replace `c138proj` with your Wi-Fi SSID, the password with your Wi-Fi password, and the 3 IP addresses with the ones you want for the TurtleBot3, the name server (DNS) and the gateway respectively.

```
sudo nano 50-cloud-init.yaml
```

In YAML files, indentation matters: use spaces (not tabs) and keep the existing indentation.

- Apply the network configuration (no reboot needed):

```
sudo netplan apply
```

The following warning is normal and can be ignored: `WARNING:root:Cannot call Open vSwitch: ovsdb-server.service is not running.`

- Troubleshooting:
 - `ip a` displays the current IP configuration of the Wi-Fi (`wlan0`) and wired (`eth0`) interfaces,
 - `sudo netplan --debug apply` displays debug information.

Method 2: directly on the SD card, from the PC (no screen or keyboard needed)

A screen and a keyboard may still be needed to debug a failed network connection.

- Insert the SD card (with the image already written) into the Ubuntu PC. Its partitions are mounted automatically, usually in `/media/<your_PC_user>/`.
- In the `writable` partition, edit the file `etc/netplan/50-cloud-init.yaml` as detailed in [method 1](#) (copy the DHCP or static configuration, then edit the SSID, password and IP addresses):

```
cd /media/$USER/writable/etc/netplan
```

then use the same `sudo cp` and `sudo nano` commands as in method 1.

- Eject the SD card, insert it into the TurtleBot3 and power it on.

Connect remotely to the TurtleBot3

Connect from a terminal on the PC to the TurtleBot3 through the network, as described in [quick_start_guide/bringup](#):

1. On the PC, check that the TurtleBot3 RPi is online:

```
ping 192.168.1.45
```

Replace `192.168.1.45` with the static IP address you gave to the TurtleBot3, or with the dynamic IP address assigned by your router/hotspot (it can be found in the router web interface, or with `ip a` on the TurtleBot3 with a screen connected). If the TurtleBot3 is connected to the network, the round-trip time of each packet is displayed. Stop `ping` with `Ctrl+C`.

2. On the PC, connect remotely to the TurtleBot3:

```
ssh ubuntu@192.168.1.45
```

Replace **192.168.1.45** with the TurtleBot3 IP address, and **ubuntu** with the user you created if you built the SD card yourself.

3. When prompted, enter the user password of the TurtleBot3 (**turtlebot** for the ready-to-use image).

Update the ROS 2 repository key

The signing key of the ROS 2 package repositories changes from time to time. If **sudo apt update** reports a **NO_PUBKEY** or **EXPKEYSIG** error for **packages.ros.org**, update the key on the RPi with:

```
sudo curl -sSL
https://raw.githubusercontent.com/ros/rosdistro/master/ros.key -o
/usr/share/keyrings/ros-archive-keyring.gpg
sudo apt update
```

OpenCR firmware update (if required)

- The latest firmware (at the time the image was built) and an update script are already installed on the SD card.
 - Run the OpenCR firmware update on the RPi:

```
cd ~/opencr
bash update_opencr.bash
```

Note: the script needs an Internet connection to install software packages.

- When asked which daemons to restart, just select *OK*.
- About 20 seconds after the end of the update, the OpenCR board should play a short melody to confirm the update.
- If you have any trouble, see the official page: [quick_start_guide/opencr_setup](#).

.bashrc

The **~/ .bashrc** file of the TurtleBot3 is already configured as expected by the manual:

```
source /opt/ros/humble/setup.bash
source ~/turtlebot3_ws/install/setup.bash
export ROS_DOMAIN_ID=30 #TURTLEBOT3
export LDS_MODEL=LDS-01
export TURTLEBOT3_MODEL=burger
```

If your LiDAR model is not **LDS-01**, replace it with **LDS-02** or **LDS-03**. Then reload the file with **source ~/ .bashrc** (or open a new terminal).

Important: the PC must use the same `ROS_DOMAIN_ID` (30) as the TurtleBot3, otherwise they cannot communicate. The PC setup of the manual adds `export ROS_DOMAIN_ID=30` `#TURTLEBOT3` to the `~/.bashrc` of the PC.

Option B: step by step with the Robotis manual

- Install Ubuntu and ROS 2 Humble on the SBC, choosing the `v4l2_camera` package method for the RPi camera:
 - [quick_start_guide/sbc_setup](#) (see the *Raspberry Pi Camera* section),
 - if `rqt_image_view` is not available when testing the camera, use `rqt -s image_view`.
- Install the OpenCR firmware:
 - [quick_start_guide/opencr_setup](#).

3. Back up the SD card

It is highly recommended to back up the content of your SD card once the installation is complete. See how to do it in [BackupRestore_SDCard.md](#).

4. Basic TurtleBot3 tests

Test the communication between the PC and the TurtleBot3, and the ROS 2 environment, with the basic teleoperation of the manual:

- [Bringup](#)
- [Teleoperation \(keyboard\)](#)

Tip: to avoid typing `export TURTLEBOT3_MODEL=burger` in every new terminal on the PC, add this line at the end of the PC `~/.bashrc` file, as on the TurtleBot3 ([quick_start_guide/export_turtlebot3_model](#)).

Troubleshooting:

- check that the PC and the TurtleBot3 **both** use the same `ROS_DOMAIN_ID` (run this in a terminal on each of them):

```
echo $ROS_DOMAIN_ID
```

- check that the PC and the TurtleBot3 are on the same network (see [Connect remotely to the TurtleBot3](#)),
- check that the motors and the OpenCR board work correctly: [OpenCR test](#),
- reboot the TurtleBot3.